

AI/ML Task 2: Model Comparison Report

California Housing Price Prediction

Methodology

The California Housing dataset from scikit-learn was used for this project. After loading the dataset, exploratory data analysis (EDA) was performed using dataset information, summary statistics, and null-value checks. The target variable was house price. The data was split into training (80%) and testing (20%) sets. StandardScaler was applied to Linear Regression and Ridge Regression models. Three machine learning algorithms were trained and evaluated: Linear Regression, Ridge Regression, and Decision Tree Regressor. Performance was measured using RMSE and R^2 Score.

Results

Model Comparison Table

Linear Regression — RMSE: 0.745581, R^2 Score: 0.575788
Ridge Regression — RMSE: 0.745557, R^2 Score: 0.575816
Decision Tree Regressor — RMSE: 0.644568, R^2 Score: 0.682948

The Decision Tree Regressor achieved the best performance among the three models. It produced the lowest RMSE (0.644568), indicating smaller prediction errors, and the highest R^2 Score (0.682948), meaning it explained approximately 68.3% of the variance in housing prices. Linear Regression and Ridge Regression showed nearly identical performance with R^2 scores around 57.6%.

Conclusion

Based on the evaluation metrics, the Decision Tree Regressor is the best-performing model for this dataset. Its ability to capture non-linear relationships helped it outperform Linear Regression and Ridge Regression. Future improvements could include hyperparameter tuning, Random Forest Regression, Gradient Boosting, and cross-validation techniques to further improve predictive accuracy and model robustness.